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**Seminar in Topics in Labor Economics and the Economics of
Education**

Did Remote Work Narrow the Motherhood Penalty?

**Evidence on the Intensive Margin from Israeli Labor Force
Survey Microdata, 2017–2023**

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Abstract

Harrington and Kahn (2025) ask whether the post-pandemic rise of work from home (WFH) reduced the motherhood penalty in the United States. This paper takes that question to Israel and to a different margin of labor supply. Using Central Bureau of Statistics (CBS) Labor Force Survey microdata on women aged 25–59 in 2017–2019 and 2021–2023, we estimate difference-in-differences (DiD) and triple-differences (DDD) models of employment and of usual weekly work hours for mothers relative to childless women, using the COVID-19-driven shift to remote work as the “post” period and a WFH-exposure index as the third difference (occupation-level for the hours model, demographic-cell-based for the employment model). On the extensive margin, the mother–non-mother employment gap did not change ($\text{Mother} \times \text{Post} = -0.0053$, SE 0.0052), and the employment DDD is underpowered rather than informative (its minimum detectable effect is roughly eight to ten times its own point estimate). On the intensive margin, employed mothers’ weekly hours did not change on average either ($\text{Mother} \times \text{Post} = 0.23$, SE 0.18). The effect is instead *confined* to teleworkable work: the triple interaction with WFH exposure is 3.22 hours per unit of exposure (SE 1.02, $p < 0.01$), while the hours change for mothers at zero exposure is if anything negative. That estimate is robust to a generalized trimming-bounds correction for selection into the estimation sample, with a 95% confidence interval for the identified set of $[1.02, 5.32]$, and fathers show no corresponding exposure gradient. We read this as WFH access relaxing a binding time constraint for mothers who remain employed in teleworkable jobs — and, because there is no average effect to speak of, as a result about *which* mothers benefit rather than about mothers as a group.

1 Introduction

Harrington and Kahn (2025) pose a simple question for the United States: has the rise of work from home reduced the motherhood penalty in the labor market? Their answer is yes on the extensive margin and on income — a 10% rise in WFH raises mothers’ employment by roughly 0.78 percentage points relative to other women, concentrated in careers with high returns to hours and inflexible time demands, and employed mothers’ incomes rise by about 1.3% relative to other employed women. This paper asks the same question of a different country and a different margin. We use Israeli CBS Labor Force Survey microdata to test whether the post-COVID shift to remote work altered the motherhood penalty in Israel, and our headline result concerns *usual weekly work hours* among employed women rather than employment or income. It is therefore an extension of Harrington and Kahn’s design to another labor market and to the intensive margin, not a direct replication of their US finding: we also estimate the employment margin, but — for reasons of statistical power that we document rather than paper over — that

specification is secondary.

The question carries both scientific and policy significance. Scientifically, the sudden, large, and plausibly exogenous shift in the location of work constitutes a rare natural experiment in workplace flexibility. This is the very margin that Goldin (2014) identifies as the last major driver of the remaining gender pay gap in developed labor markets. Temporal and locational flexibility is a good mothers may pay for, in the form of lower wages or slower promotion. A shock that lowers the cost of supplying flexibility should, in principle, be detectable in mothers’ labor-supply choices — and, for mothers already employed, most directly in how many hours they supply. Unlike the acute, transitory disruption of 2020, WFH has proven to be a persistent feature of the post-pandemic labor market rather than a temporary accommodation (Barrero et al., 2021). This renders the question policy-relevant rather than merely historical.

The policy stakes are particularly high in Israel, which combines a highly productive, internationally integrated labor market with the OECD’s highest total fertility rate. A labor market that induces mothers of young children to reduce employment or hours imposes a substantial aggregate cost. This cost is especially large in a country where a large and growing share of the working-age population is, at any given time, a parent of young children. Conversely, flexible work arrangements might measurably narrow the gap between mothers and other workers. If so, the finding bears directly on the design of remote-work policy, parental-leave rules, and public discourse on gender equality in the labor market. At the same time, Israeli institutional practice diverges sharply from US task-based predictions of which jobs can be done remotely — schools stayed in person by policy, for example — which matters for how “WFH exposure” should be measured and is one of the methodological contributions below.

Our research design exploits the pandemic as the moment at which remote work became a mainstream labor-market practice. Before 2020, working from home accounted for roughly 5% of workdays; after the pandemic it accounted for 25–30%. We therefore compare mothers and childless women across a pre-period of 2017–2019 and a post-period of 2021–2023, excluding 2020 as a transitional year of emergency, ad hoc arrangements that reflect improvisation rather than the settled remote-work practices whose effect we want to measure. (A second, practical reason is that no 2020 CBS extract exists for this project at all; Section 3 discusses what that implies for the exposure index.) The “post” dummy captures this structural shift toward remote work.

The core estimator is an individual-level DiD of the form $Y_{it} = \beta_0 + \beta_1 \text{Mother}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Mother}_i \times \text{Post}_t) + X'_{it} \gamma + \varepsilon_{it}$, where β_3 is the differential post-shift change for mothers. To test whether remote work is the *mechanism*, we extend this to a triple-differences (DDD) model that interacts $\text{Mother} \times \text{Post}$ with an occupation’s WFH exposure, so that the parameter of interest is whether the narrowing of the penalty is larger in jobs that can actually be done from home.

Our findings are as follows. On the extensive margin there is nothing to see and, more importantly, nothing the design could have seen: the employment DiD is a precise zero, and the employment DDD has a minimum detectable effect roughly 8–10 times its point estimate. On the intensive margin, mothers’ hours did not rise on average either: the plain Mother $\times$ Post term is 0.23 (SE 0.18), a reasonably precise null that rules out average gains above about half an hour a week. The result is that there is no average effect to decompose — the entire effect lives in the interaction with occupational WFH exposure, 3.22 additional weekly hours per unit of exposure ($p < 0.01$), while the implied change for mothers at zero exposure is if anything negative. This estimate survives a selection correction, two age-balance robustness specifications, and a gender placebo: fathers show no positive exposure gradient, their triple interaction being negative and significant only at the 10% level. We are careful about what the placebo does and does not establish on the plain DiD, where fathers move *more* than mothers do; we return to this in Section 6.4.

The remainder of the paper proceeds as follows. Section 2 places the question in the literature. Section 3 describes the data, the sample, and the construction of the WFH exposure measures. Section 4 describes the analysis sample and shows both margins and the exposure measure before any model is imposed on them, and presents the commuting evidence bearing on the mechanism. Section 5 states the hypotheses and sets out the DiD and DDD specifications and the selection correction. Section 6 reports the results, moving from the extensive to the intensive margin. Sections 7–8 interpret the findings and state their limitations, and Section 9 concludes.

2 Literature Review

The motherhood penalty. The child penalty is one of the most robust facts in labor economics. Kleven et al. (2019b) use Danish administrative data and an event-study design around the birth of the first child to show that women’s earnings fall by roughly 20% in the long run relative to men’s, with the gap fueled by reduced work hours, exit from the labor market, and moves into family-friendly, lower-paying jobs. Kleven et al. (2019a) extend the event-study approach across countries and find that the qualitative pattern is universal while its magnitude varies sharply across countries, and argue that differences in gender norms, rather than in family policy, are the more likely explanation. The Danish study matters here in particular because it identifies *hours* as one of the channels through which the penalty operates — which is why an intervention that changes the time cost of combining work and care could plausibly move the intensive margin even where it does not move employment.

Temporal flexibility as the mechanism. Goldin (2014) argues that the last chapter of gender convergence turns on the structure of jobs rather than on human capital: occu-

pations that reward long, particular, and inflexible hours — where the marginal return to being physically present at a specific time is high — carry the largest gender gaps, and the gap would shrink if firms did not disproportionately reward temporal inflexibility. Workers who need temporal flexibility — disproportionately mothers — pay a wage penalty for it. This framework generates a sharp, testable prediction: an exogenous change that lowers the cost of supplying flexibility should differentially help mothers, either by raising their employment probability, their hours, or both. Remote work is, in this reading, precisely such a change in temporal flexibility. It lowers commute friction, allows a virtual presence in demanding professions, and may reduce the stigma attached to a mother’s physical availability. It may also cut the other way, through lower visibility and slower advancement. Which effect dominates is an empirical question, and Goldin’s framework tells us where to look: in occupations where the work *can* be done remotely.

Does remote work narrow the gap in practice? The evidence on whether flexible-work arrangements realize this potential is mixed. In a field experiment on Italian white-collar workers, [Angelici and Profeta \(2024\)](#) find that randomly assigned flexible (“smart”) working arrangements shifted the division of household tasks. Women’s reported well-being also increased, suggestive of a genuine flexibility benefit. Real-time survey evidence from the acute pandemic period tells a more ambiguous story. [Adams-Prassl et al. \(2020\)](#) show that WFH-feasible jobs were substantially better shielded from the 2020 employment and earnings shock. Women in WFH-feasible jobs, however, simultaneously absorbed a larger share of the additional childcare burden created by school and daycare closures. The same shock that protected employment could, in principle, degrade the quality of that employment for mothers specifically. [Alon et al. \(2020\)](#) formalize the macro logic for why COVID-19-type shocks hit mothers asymmetrically (sector composition plus school closures). They do not take a stand on whether WFH exposure specifically offsets or compounds this asymmetry. This paper is designed to adjudicate empirically between these two directions using post-acute-period (2021–2023) data, rather than the initial-shock period that most of this literature studies.

Measuring who can work from home. [Dingel and Neiman \(2020\)](#) classify occupations by whether their tasks can be performed entirely at home, using O*NET task content, and find that 37% of US jobs can be done from home. Their occupational classification, mapped to the two-digit ISCO-08 codes in the CBS data, is the starting point for our WFH-exposure index. As Section 3 explains, however, a US task-based measure misclassifies occupations where Israeli institutional practice diverged from what the tasks would predict, and we therefore calibrate it against realized Israeli behavior. For the Israeli labor market specifically, [Yaish et al. \(2021\)](#) document that the 2020 lockdown reallocated paid and unpaid labor within dual-earner households in gender-unequal ways.

Madhala and Bental (2020) provide a descriptive occupation-level baseline of WFH feasibility among Israeli workers. Buzaglo-Baris (2023) shows that within-firm sorting, not occupation choice alone, is a first-order driver of the Israeli gender wage gap. This is a reminder that any occupation-based exposure measure captures only part of the relevant heterogeneity.

Remote work and the motherhood penalty. The paper closest to ours is Harrington and Kahn (2025), who show for the United States that the rise of WFH raised mothers’ employment relative to other women’s, that this is driven by fields with high returns to hours and inflexible demands on time, and that women who could work from home before childbirth are less likely to exit the workforce. We take their question to Israel. Our contribution is threefold: a different country with a distinctive combination of high fertility and high productivity; a different and, in the child-penalty literature, central margin — hours conditional on working — with an explicit correction for the selection problem that conditioning on employment creates; and an exposure measure calibrated to Israeli practice rather than imported from US task content. Because the data span both pre- and multiple post-COVID years, they also allow separate identification of extensive- and intensive-margin responses — a distinction the WFH-and-gender literature above largely does not draw. And because the exposure measure is built from pre-determined occupational content rather than realized WFH status, it avoids the reverse-causality problem inherent in treating self-reported WFH as an explanatory variable for hours.

Identification. The triple-differences estimator we use for the mechanism test is analyzed formally by Olden and Møen (2022), who clarify that the DDD does not require two parallel-trends assumptions but rather a single assumption that the *difference* between the two DiDs would have evolved in parallel absent treatment. Section 5 states the identifying assumptions in those terms.

3 Data and Institutional Context

3.1 Source and sample

The data are individual-level microdata from the Israeli Central Bureau of Statistics Labor Force Survey for the survey years 2017, 2018, 2019, 2021, 2022, and 2023. The analysis sample is women aged 25–59 (CBS age-group codes 3–7). A woman is coded as a *mother* if she has any child under 17 in the household, and as childless otherwise; the treatment group is mothers and the control group is childless women. *Post* equals one for survey years 2021 and later.

Two outcomes are studied. The primary outcome is *usual weekly work hours*, a continuous variable constructed from the CBS binned hours item by assigning each bin its median and imputing irregular-hours codes from observed medians within the same period; it is defined for women who are employed *and* worked in the survey’s reference week.

That second restriction is not cosmetic, and it is worth stating plainly because it changes the estimand. In the 2017 file, respondents who were employed but absent from work in the reference week had their usual hours recorded as zero — the code CBS otherwise uses for “no usual hours” — whereas from 2018 onward the same respondents were assigned a real usual-hours value. Zero is a defensible entry for *actual* hours in a week not worked; it is not a usual-hours value, and left uncorrected it enters the pre-period as roughly 5,200 spurious zeros. Because absence from the reference week is itself more common among mothers (12.4% of employed mothers against 7.3% of childless women), those zeros do not fall evenly across the comparison this paper makes. We therefore define hours on the reference-week workers in *every* year, which costs about 10% of each year’s employed sample and makes the hours population identical across years. The estimand is correspondingly narrower: usual hours among women who were working, not among all women who held a job. Section 8 returns to what this does and does not fix.

The secondary outcome is a binary employment indicator, equal to one if the respondent reports being employed and zero otherwise (unemployed and not-in-labor-force are both zero). Among women working in the reference week the pooled mean of weekly hours is 39.18 ($n = 258,175$) and the pooled employment rate is 0.7737 ($n = 372,741$).

All regressions include the same set of categorical controls: marital status, religion, age group, district of residence, and highest educational attainment (six groups). Age enters as the categorical CBS age-group code because the extract contains no continuous age or birth-year variable. CBS survey weights are deliberately not applied in any outcome regression; all reported estimates are unweighted estimates on the analysis sample rather than population-representative statistics (see Section 8).

3.2 Why 2017–2019 versus 2021–2023

The window is designed around the pandemic’s role in making WFH a mainstream practice: working from home rose from about 5% of workdays before 2020 to 25–30% afterward. The post-period is therefore 2021–2023 and the pre-period 2017–2019. The year 2020 is excluded as a transitional year. There is also a data-driven reason: no 2020 CBS extract was acquired for this project, so 2020 could not be included even in principle. The consequence that matters for the design is discussed next.

3.3 Constructing WFH exposure

Since realized WFH status is itself a labor-market outcome — jointly determined with employment and hours — causal designs need a measure of WFH *exposure* that is not chosen by the individual whose hours are being modeled. The mechanism test therefore requires an occupation-level measure of how exposed each job is to remote work. We build it in three steps, each producing a measure that is reported.

External. The [Dingel and Neiman \(2020\)](#) teleworkability score, via an O*NET/SOC-to-ISCO crosswalk, at the two-digit ISCO-08 level. This is pre-period by construction and immune to Israel’s own COVID-era behavior, but it is a US task-based measure and misclassifies occupations where Israeli institutional practice diverged: teaching (ISCO 23) scores 0.966 on the external index although Israeli schools stayed in person by Ministry of Education policy.

Calibrated (primary). We correct the external score only where the gap between it and the realized 2022–23 Israeli WFH share is large (greater than 0.5) *and* statistically distinguishable from sampling noise, using a one-sided cluster-robust test at the 95% level. Ten of the 40 two-digit occupations are swapped to their realized Israeli values. The corrections are concrete and checkable: teaching falls from 0.966 to 0.069, clerical support (ISCO 41) from 1.000 to 0.098, while ICT professionals (ISCO 25) go from 1.000 to 0.360. Because it draws on 2022–23 realized data, which sit inside the post-period, the calibrated measure is a documented compromise rather than a fully pre-treatment index; the exposure measures are built from a population that includes men as well as the women in the analysis sample, so that an occupation’s score is not constructed from exactly the people who enter the regression.

Realized. The realized Israeli WFH share by occupation, anchored to 2021 with a minimum of 200 observations per occupation. This is post-treatment by construction and is reported as a robustness check.

The original research plan specified anchoring the realized index to the 2020 WFH variable. Because no 2020 extract exists, the index is anchored to 2021 instead. This is a disclosed deviation with a stated justification — 2021 already flows through the validated pipeline, and a measure taken during 2020’s emergency disruption might reflect improvisation rather than each occupation’s settled remote-work potential — and a stated downside: by 2021 occupations had had a year to adapt, and if adaptation was uneven a 2021-based index could rank occupations differently than a 2020-based one would have. We return to this in Section 8.

For the employment-outcome DDD, occupation is undefined for the non-employed, so an occupation-level regressor would condition the outcome on itself. That specification therefore uses a fourth measure: a pre-period (2017–2019) shift-share index by demographic cell (sex, age group, education, district, marital status, religion, and continent

of birth — 8,884 cells), equal to the weighted mean of the calibrated occupation score over the cell’s pre-period occupational composition. It is defined for employed and non-employed women alike. The two DDDs thus use *different* exposure measures, and we are careful never to compare their coefficients as if they were on a common scale.

4 Descriptive Statistics

This section sets out what the analysis sample looks like before any model is imposed on it. The order is deliberate: the summary table, then the primary (hours) margin, then the exposure measure that identifies the triple difference, then the commuting evidence bearing on the mechanism, and finally the secondary (employment) margin. Nothing here is adjusted for the controls listed in Section 3, and nothing here carries an identifying assumption; the estimates begin in Section 6.

One convention applies throughout. All standard errors and confidence intervals in this section are clustered by individual, as they are in Section 6. This matters more than it may appear: the survey is a rotating panel in which a respondent is observed roughly four times overall and two to three times within a single year, so the independent-observations formulas normally used for a cell mean or a difference of proportions understate these standard errors by a factor of about 1.8 to 2. The point estimates are ordinary cell arithmetic — a difference of means is still a difference of means — but the intervals around them are not.

4.1 The sample in summary

Table 1 describes the sample on either side of the treatment/control split. Two features matter for what follows. Mothers are *more* likely to be employed than childless women (0.7828 versus 0.7574) but work *fewer* hours when they do (38.68 versus 40.06), so the *level* of the penalty is visible in the raw data on the intensive margin and runs the other way on the extensive one. Whether that level changed after 2021 — which is the question this paper actually asks — is not something Table 1 can show. And mothers are the better-educated group, with 47.97% holding an academic degree against 40.24% of childless women, which is why the controls matter: the unconditional hours gap understates the conditional one. The largest single difference is marital status: 84.57% of mothers are married against 46.55% of childless women, with the single, never-married share running the other way (5.90% versus 34.00%), so marital status does real work as a control. The two groups are also imbalanced on age, mothers sitting about 0.44 of an age-group code younger, an imbalance Section 6 returns to because it tracks WFH exposure. Religion and district are close to balanced, the one visible tilt being that mothers are more concentrated in Jerusalem and Judea and Samaria and less in Tel Aviv.

Table 1: Summary statistics: women aged 25–59, by mother status

	Childless women	Mothers	Difference
<i>Panel A: outcomes</i>			
Observations	133,646	239,095	
Employment rate	0.7574	0.7828	0.0254
Usual weekly hours (reference-week workers), mean	40.06	38.68	−1.38
standard deviation	[12.12]	[11.62]	
<i>Panel B: characteristics</i>			
Age-group code (CBS 3–7), mean	5.40	4.96	−0.44
Arab (%)	17.73	17.78	0.05
<i>Marital status (%)</i>			
Married	46.55	84.57	38.02
Married, living separately	3.58	0.84	−2.74
Divorced	13.13	7.70	−5.42
Widowed	2.75	0.99	−1.76
Single, never married	34.00	5.90	−28.10
<i>Religion of household head (%)</i>			
Jewish	73.08	79.50	6.42
Christian	7.12	2.23	−4.89
Muslim	13.78	14.89	1.11
Druze	1.88	1.64	−0.24
Other	4.15	1.75	−2.40
<i>District of residence (%)</i>			
Jerusalem	8.76	12.31	3.56
North	17.52	16.58	−0.94
Haifa	14.38	11.92	−2.45
Center	23.99	26.52	2.53
Tel Aviv	21.55	14.90	−6.65
South	11.66	12.37	0.72
Judea and Samaria	2.15	5.39	3.24
<i>Education (%)</i>			
Below high school	11.37	6.63	−4.74
High school, no matriculation	12.70	11.95	−0.74
Matriculation (Bagrut)	22.73	18.82	−3.90
Post-secondary, non-academic	11.50	13.68	2.18
Academic degree (BA/MA/PhD)	40.24	47.97	7.73
Other / no certificate	1.47	0.95	−0.52

Israeli CBS Labor Force Survey, survey years 2017–2019 and 2021–2023, women aged 25–59. A woman is a mother if she has any child under 17 in the household. Weekly hours are conditional on working in the reference week (Section 3); the standard deviation is in brackets. Within each categorical block, percentages sum to 100 in each column. The difference column is mothers minus childless women. All figures are *unweighted* counts on the analysis sample and are not population-representative (Section 8). The Arab share is the CBS population-group code, the variable used for the subgroup analysis; religion is a separate CBS item recorded for the household head. Level labels for marital status, religion and district follow the CBS codebook that accompanies the extract. Marital status, religion, district, age group and education are the five controls in every regression.

4.2 The raw hours gap

Before any model, the primary outcome in four cell means. Mothers work 38.54 hours per week before 2021 and 38.84 after, a gain of 0.31 hours. Childless women move from 40.04 to 40.08, a gain of 0.04. The difference between those two changes is 0.266 hours, and it is the entire difference-in-differences of equation (1) computed without a single control variable.

This matters for how the later estimate should be read. The regression DiD of Table 5 is 0.2280 (SE 0.1809), within four hundredths of an hour of the raw figure and with an almost identical standard error. What the controls do not do, in other words, is manufacture or conceal an average effect: the near-absence of one is already present in the arithmetic of four averages, and neither version is distinguishable from zero.

Note also that mothers remain well below childless women in both periods, and that whatever movement there is would close only a small fraction of that gap even if one took it at face value. The average mother’s hours are not what moves in this paper; Section 4.4 shows what does.

4.3 Hours year by year

Pooling the pre-period into a single average hides whether the two groups were on a common trajectory before 2021, so Figure 1 un-pools it. The lower panel plots the raw mother-minus-non-mother hours gap in each survey year. The pre-period gap is -1.37 hours in 2017, -1.48 in 2018 and -1.67 in 2019: three years within about a third of an hour of each other, and each within the others’ confidence bounds. Whatever else is true of this design, the groups were not diverging before treatment.

This is the descriptive counterpart of the formal test in Table 4, which does not reject ($F(2, 63, 198) = 0.949$, $p = 0.387$). It is worth being explicit that this was not always so. Before the hours population was harmonized on reference-week work (Section 3), the 2017 gap read -3.30 and the pre-trend test rejected decisively — an artifact of the spurious zeros described there, which fell disproportionately on mothers and so manufactured exactly the kind of pre-existing divergence a reader should worry about.

The same panel shows the post-period path: the gap is -1.37 in 2021, widens slightly to -1.53 in 2022, and then narrows to -0.80 in 2023, the smallest value in the sample. The movement is not monotone, and at these standard errors the three post-period years are not individually distinguishable from one another; it is the event study of Section 6.3, not this raw series, that supports the gradual-adjustment reading in Section 7.

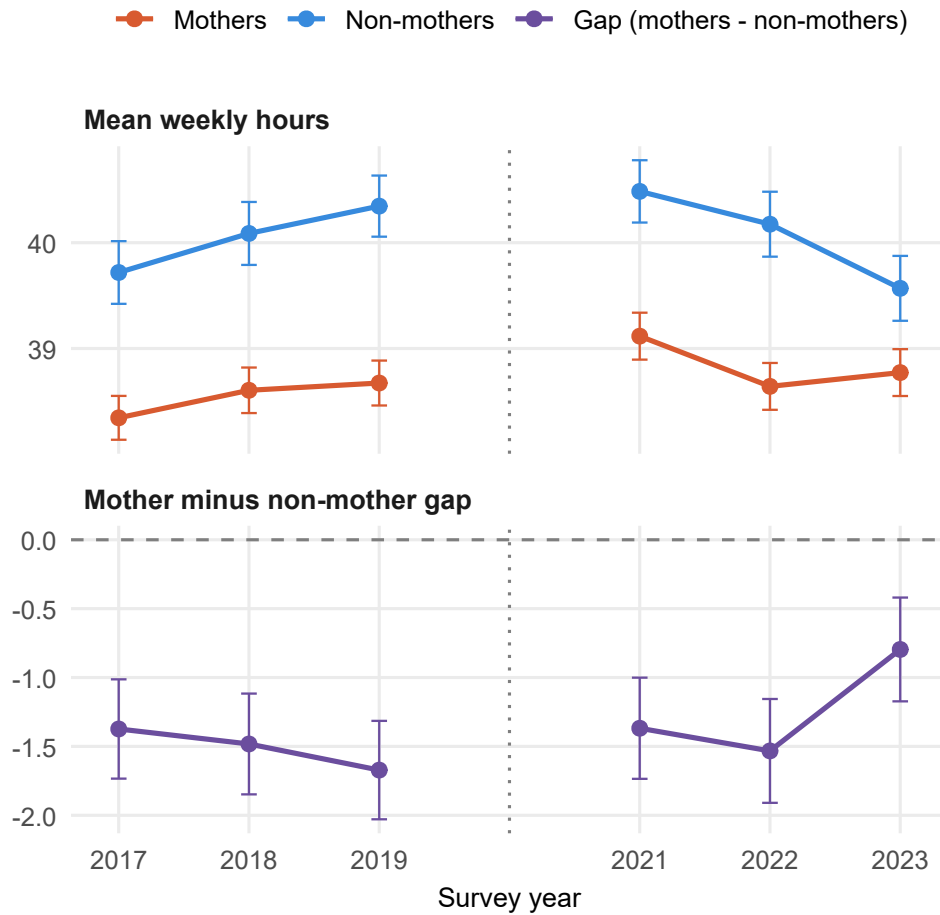


Figure 1: Weekly hours by survey year: levels and the motherhood gap
Notes: Women aged 25–59 who worked in the reference week, unweighted means with no controls. The upper panel plots each group’s mean weekly hours; the lower panel plots their difference. Vertical bars are 95% confidence intervals. The dotted rule marks the excluded 2020 survey year, and the series break there rather than interpolating across it. The three pre-period gaps are the raw counterpart of the insignificant $\text{Mother} \times \text{year}$ coefficients in Table 4.

4.4 Exposure and the commuting channel

The triple difference asks whether hours respond where work can be done from home. Figure 2 puts that question to the data directly, by computing the same difference-in-differences as Section 4.2 separately within each quartile of occupational WFH exposure. The answer is not that the response is larger at the top; it is that there is no response anywhere else. The bottom three quartiles return -0.32 , 0.27 and -0.02 hours, none of them distinguishable from zero. The top quartile returns 1.55 hours (95% CI $[0.74, 2.36]$).

Two things follow. First, this is why Section 4.2’s average is near zero: it is a weighted average of three nulls and one substantial positive difference, and the three nulls dominate by construction. The decomposition is approximate rather than exact — these quartiles additionally require a matched occupation code, so they rest on 251,968 rows against Section 4.2’s 258,175, and the quartile-weighted average is 0.29 rather than 0.27. Second, the dose-response that the DDD coefficient of 3.224 summarises as a single slope is visible without any regression — but it is emphatically not linear in exposure, and the identifying contrast is a comparison of the most teleworkable quartile against the rest rather than a smooth gradient. The quartile boundaries make the same point: the bottom three quartiles span exposure values of 0.00 to 0.30, while the top quartile spans 0.30 to 0.75. Most of the dispersion in the regressor lives in one tail, which is worth bearing in mind when reading a coefficient expressed per unit of exposure.

Figure 3 supplies a descriptive reason to think commuting is part of the story. It plots the share of each group working outside their locality of residence. Before the pandemic both groups were drifting upward. In 2021 the share collapses for childless women, from 46.1% in 2019 to 42.2%, and recovers to 46.0% by 2023; for mothers the same series moves from 48.2% to 46.7% and back to 48.4% — a disruption less than half as large. Mothers were the more mobile group throughout, which is not what a simple story about mothers being tied to home would predict, and their commuting behaviour was the less disturbed of the two. This is consistent with the mechanism of Section 7, in which remote work relaxes a time constraint for women who were already commuting, rather than drawing a previously home-bound group into work.

4.5 The extensive margin by child age

Figures 4 and 5 describe the secondary outcome. Employment among mothers traces a shallow inverted U in the age of the youngest child, from 75.9% at ages 0–1 to a peak of 80.2% at ages 5–9 and back to 76.7% at ages 15–17. The pre- and post-2021 profiles are close where the youngest child is under five — 0.3 and 0.5 percentage points apart in the 0–1 and 2–4 bins — and separate by around four points in each of the three older bins. Whether that separation means anything is a question this figure cannot answer, for reasons given below.

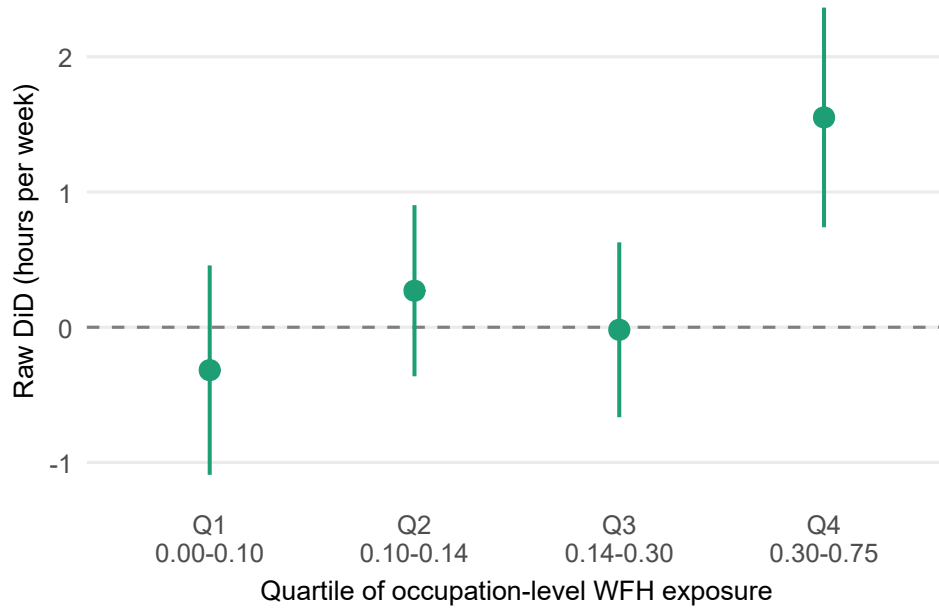


Figure 2: Raw hours difference-in-differences by quartile of WFH exposure

Notes: Women aged 25–59 who worked in the reference week and carry a matched occupation code. Each point is the (mothers post – mothers pre) – (childless post – childless pre) difference within one exposure quartile, from unweighted cell means with no controls; bars are 95% confidence intervals clustered by individual. Quartiles are of the *occupation-level calibrated* exposure measure — the same regressor the hours DDD uses, not the demographic-cell index of Section 6.2, whose scale is different and whose coefficients are not comparable. Breakpoints are computed on pre-period rows and applied to all years.

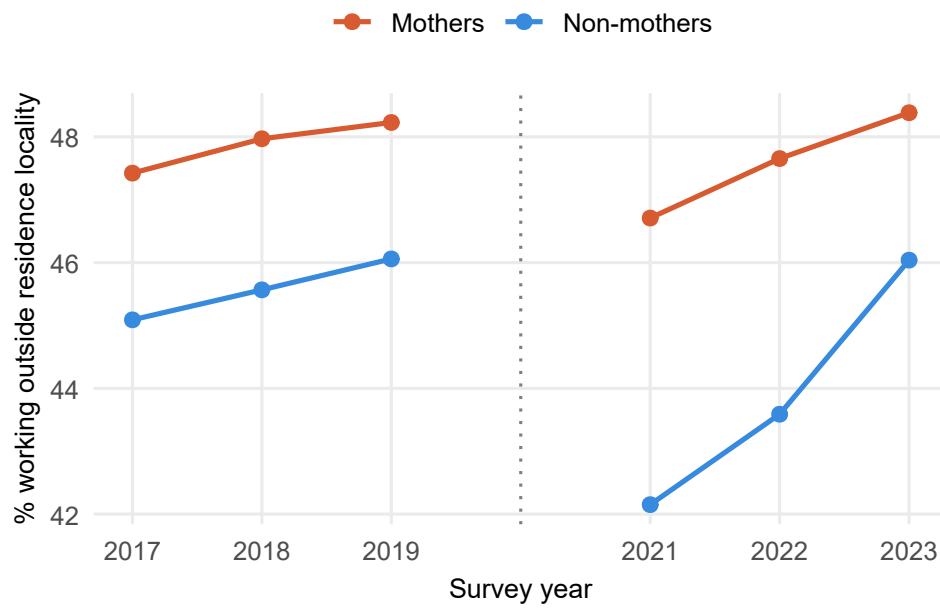


Figure 3: Share working outside the locality of residence, by year
Notes: Women aged 25–59 with a recorded locality of work, unweighted shares with no controls. That population is *broadier* than the reference-week workers the hours figures use — 56,323 against 47,728 in 2017 — so this figure is not drawn on the same sample as the rest of the section. The dotted rule marks the excluded 2020 survey year and the series break there. This figure is descriptive evidence on a proposed mechanism, not a test of it: no specification in this paper uses locality of work as an outcome.

Figure 5 shows that the downward right-hand tail is largely compositional. Holding the controls at their sample means, the adjusted profile does not turn down after age 5–9: it continues to rise, reaching 81.8% in the 15–17 bin against a raw 76.7%. Mothers whose youngest child is nearly grown are themselves older and differently educated, and once that is accounted for their employment rate is the highest in the sample rather than among the lowest.

A caution about what these two figures do and do not show. Both plot employment *levels* among mothers. Neither contains a comparison group — there are no childless women in either figure — so the visible pre/post distance is a difference in levels within one group, not an estimate of anything, and in particular not a difference-in-differences. The four-point separation in the older bins is therefore not evidence that employment rose for those mothers relative to anyone. It should not be read against the employment DiD of Table 2, which is estimated relative to childless women and with controls, and which is a precise null.

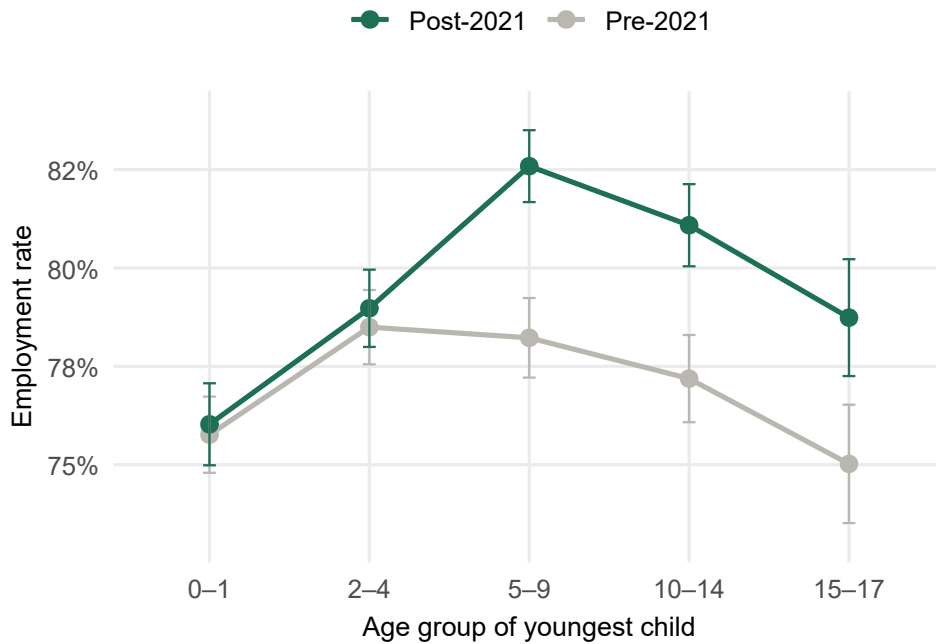


Figure 4: Employment rate by age of the youngest child, pre and post 2021

Notes: Mothers aged 25–59, unweighted employment rates with no controls; bars are 95% confidence intervals clustered by individual, which are roughly twice the width of the binomial intervals the independence assumption would give. Mothers only — there is no comparison group in this figure, so the distance between the two profiles is a description of levels and not an estimated effect.

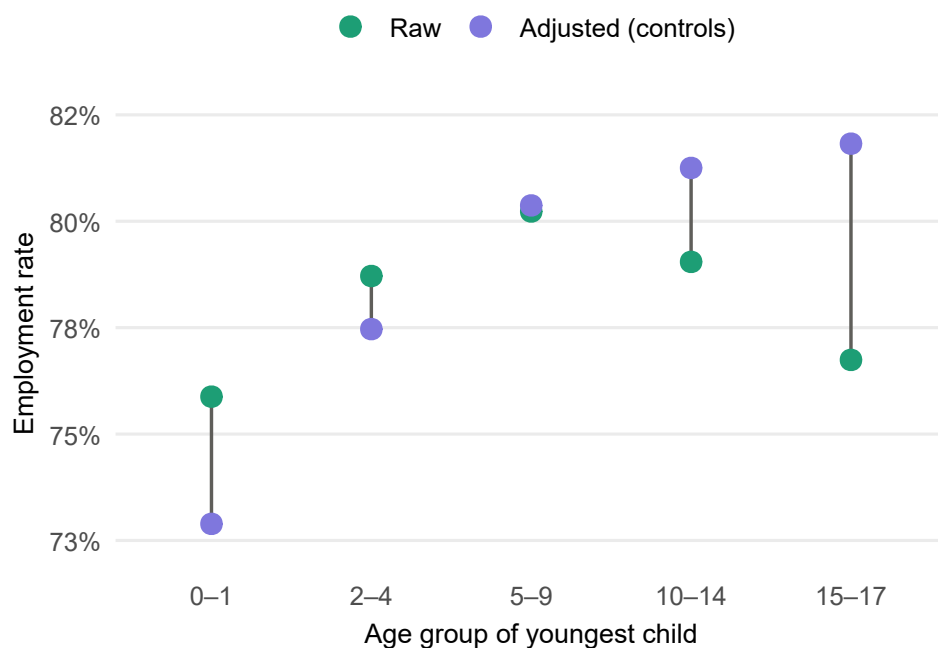


Figure 5: Raw versus covariate-adjusted employment rate by age of the youngest child

Notes: Mothers aged 25–59. Adjusted rates are OLS predictions from a regression of employment on the youngest-child age bin and the standard controls, averaged over the sample with the controls held at their means. Segments join each bin’s raw and adjusted rate. The vertical axis does not begin at zero. As in Figure 4, this is mothers only: the raw-to-adjusted distance is a statement about composition within that group, not an effect.

5 Empirical Strategy

5.1 Hypotheses

Two hypotheses follow directly from the flexibility framework of [Goldin \(2014\)](#) and the mixed empirical record reviewed above. **H1 (intensive margin):** WFH exposure raises mothers' hours conditional on working relative to childless women after 2021, if flexibility relaxes a binding time constraint while the participation margin remains the more costly one to adjust. **H2 (extensive margin):** occupational WFH exposure raises mothers' employment probability relative to other women after 2021, if flexibility relaxes a binding participation constraint. H1 is the paper's primary hypothesis; H2 is tested but, for the power reasons documented below, is secondary.

5.2 Difference-in-differences

For outcome Y_{it} (employment, or weekly hours conditional on working that week) we estimate

$$Y_{it} = \beta_0 + \beta_1 \text{Mother}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Mother}_i \times \text{Post}_t) + X'_{it} \gamma + \varepsilon_{it}, \quad (1)$$

where X_{it} is the control vector described above and standard errors are clustered by individual identifier. β_3 is the DiD estimator: the differential change in the outcome for mothers relative to childless women after the shift to remote work. For the hours specification a positive β_3 indicates a narrowing of the penalty on the intensive margin; for employment it indicates a narrowing on the extensive margin. The employment DiD is also estimated separately for Jewish and Arab women.

5.3 Triple differences

To test whether the narrowing operates through remote work, we interact the DiD with occupational WFH exposure:

$$Y_{it} = \alpha + \text{Mother}_i \times \text{Post}_t \times \text{WFH_Exposure}_{j(i)} \text{ (all lower-order terms)} + X'_{it} \gamma + \varepsilon_{it}. \quad (2)$$

The coefficient on the triple interaction asks whether the post-shift change for mothers relative to non-mothers is larger in more exposed occupations. Following [Olden and Møen \(2022\)](#), the identifying assumption is that the mother/non-mother gap in high-versus low-exposure jobs would have evolved in parallel absent the shift to remote work. For the hours DDD, WFH_Exposure_j is the calibrated occupation-level score of [Section 3.3](#), the sample is employed women matched to an occupation, and standard errors are clustered by two-digit occupation — the level at which the regressor varies, so that

individual-level clustering would understate the standard error. For the employment DDD, WFH_Exposure is the cell-based shift-share index, estimated on the full sample of employed and non-employed women, with a Mother $\times$ age group interaction added to correct a documented pre-period age imbalance between mothers and non-mothers, in two variants: additive controls (Spec 1) and fully interacted age-by-education-by-district cell fixed effects (Spec 2). Standard errors are clustered on the age-by-education-by-district cell (about 210 clusters). We also report the exposure-specific minimum detectable effect, $MDE = SE \times (z_{0.975} + z_{0.80}) = 2.8016 \times SE$, so that a null coefficient can be read as “genuinely null” or “underpowered.”

5.4 Selection into employment: Lee bounds

Hours are observed only for women who are working, and whether a woman is working is itself an outcome of the same treatment. If remote work differentially pulls marginal mothers into work after 2021, the post-period sample of working mothers is compositionally different from the pre-period one for reasons unrelated to hours choices, biasing β_3 in the hours equation in a direction that is not signable a priori. We address this with trimming bounds in the spirit of Lee (2009), adapted to the two-by-two design. Let $s_{ab} = \Pr(\text{hours observed} \mid \text{Mother} = a, \text{Post} = b)$ — that is, the probability of being employed *and* working in the reference week, which is the sample the outcome is measured on rather than the employed population as a whole. The counterfactual below assumes parallel trends *in selection*. For post-period mothers it is $s_{11}^* = s_{10} + (s_{01} - s_{00})$. If the observed s_{11} exceeds s_{11}^* , the excess share $p = 1 - s_{11}^*/s_{11}$ is trimmed from the (Mother = 1, Post = 1) cell’s hours distribution — from the top to obtain a lower bound on β_3 , from the bottom to obtain an upper bound — and equation (1) is re-estimated on each trimmed sample. For the DDD we generalize this construction: the selection counterfactual is computed separately within each quartile of the cell-based exposure index (which, unlike the occupation-level index, is observable for the non-employed), each quartile’s post-period mothers are trimmed by that quartile’s own excess share, the trimmed subsets are recombined, and equation (2) is refit with the occupation-level regressor. Two different exposure measures thus serve two different roles. Because the parameter is only partially identified, we report, alongside each bound’s own cluster-robust confidence interval, the Imbens and Manski (2004) confidence interval for the identified set, which accounts jointly for the sampling uncertainty of both endpoints. The construction handles only *excess* selection in the treated cell, and assumes monotone selection; both limitations are stated in Section 8.

5.5 Diagnostics and placebo

We estimate event-study versions of equation (1) with $\text{Mother} \times \text{year}$ interactions (reference year 2019) for both outcomes and test the pre-period coefficients jointly. Because those test the difference-in-differences assumption rather than the triple-differences one, we also estimate the event-study version of equation (2) itself — Post replaced by survey-year interactions at every level, so that $\text{Mother} \times \text{year} \times \text{WFH_Exposure}$ is recovered separately for each year — and apply the same joint test to its pre-period coefficients. We re-estimate the hours DDD with an added $\text{Mother} \times \text{age-group}$ interaction and with pre-period age-group raking weights, in response to the age imbalance noted above; with the external and realized exposure indices in place of the calibrated one; and, as a gender placebo, on the male subsample with “mother” read as “father,” comparing the male and female coefficients with two-sample z -tests (valid without a covariance correction because the samples are disjoint).

6 Results

Throughout, significance is denoted *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, · $p < 0.1$. Standard errors are in parentheses.

6.1 Extensive margin: the employment DiD

Table 2 reports equation (1) with employment as the outcome, for all women and separately for Jewish and Arab women. The DiD coefficient is -0.0053 (SE 0.0052) in the pooled sample: the mother/non-mother employment gap did not change after the shift to remote work. The estimate is precise — its standard error is about half a percentage point — so this is a tight null for the employment *DiD*, not an uninformative one — a distinction that matters, because the employment DDD in Section 6.2 is underpowered rather than precise, and the two should not be read as one verdict. The stratified estimates are -0.0041 (SE 0.0058) for Jewish women and -0.0208 (SE 0.0138) for Arab women, neither significant.¹ Employment rose in the post-period for all women ($\text{Post} = 0.0153^{***}$ pooled), and markedly so for Arab women (0.0601^{***}).

¹The religion control (**Dat**) is not identified within the Jewish-only subsample and enters with fewer categories in the Arab-only subsample, so the effective control set differs slightly across the stratified models. Its smallest category (“other”) is also dropped for collinearity in two of the pooled models, this employment DiD and the hours DDD of Table 5, so those too estimate four religion categories rather than five. No reported coefficient is affected; the dropped column carries no independent variation once the other controls are in.

Table 2: Extensive-margin DiD: employment

	All women	Jewish women	Arab women
Mother $\times$ Post	−0.0053 (0.0052)	−0.0041 (0.0058)	−0.0208 (0.0138)
Mother	−0.0034 (0.0041)	0.0054 (0.0045)	0.0200 (0.0110)
Post	0.0153*** (0.0043)	0.0070 (0.0049)	0.0601*** (0.0111)
Constant	0.4926*** (0.0085)	0.5920*** (0.0119)	0.6300*** (0.0582)
Observations	364,784	283,975	65,598
R^2	0.22689	0.04336	0.24314

Dependent variable: employment indicator. Controls: marital status, religion, age group, district, education. Standard errors clustered by individual. Jewish/Arab samples defined by the CBS population-group code.

6.2 Extensive margin: the employment DDD is underpowered

Table 3 reports the employment DDD with the cell-based exposure index. The triple interaction is -0.0257 (SE 0.0725) with additive controls and -0.0194 (SE 0.0722) with interacted cell fixed effects — small, negative, and nowhere near significant. The question is whether this is evidence of no mechanism or evidence of nothing. The minimum detectable effect answers it: at $\alpha = 0.05$ and 80% power, the design could detect a triple interaction no smaller than 0.2031 (Spec 1) or 0.2022 (Spec 2), roughly 8–10 times the point estimates. That ratio is the cleanest statement of the problem, because it is invariant to how the regressor is scaled: the design cannot detect an effect the size of the one it estimates.

Putting the coefficient on an interpretable scale requires care, because it is an effect per *unit* of exposure and the cell-based index never moves a unit: on the estimation sample its standard deviation is 0.0706 and its interquartile range 0.0766. A one-standard-deviation increase in exposure would therefore have to shift mothers’ relative post-2021 employment probability by 1.43 percentage points, or 1.85% of the baseline employment rate of 0.7737, before this design could see it. For comparison, [Harrington and Kahn \(2025\)](#) report roughly 0.78 percentage points per 10% rise in WFH, so the design is underpowered against the closest available benchmark by a factor of about two to three. The null here is therefore *uninformative*: the employment DDD cannot be read as evidence for or against a WFH-mediated employment effect.

The shortfall is structural rather than a specification choice, and we report that having tried to close it. Clustering on the exposure cell rather than the coarser fixed-effect cell changes the standard error from 0.0725 to 0.0711; estimating the combined margin instead,

with hours coded zero for the non-employed, leaves the triple interaction at less than half its own minimum detectable effect; and two successive refinements of the exposure-cell partition had already cut the detectable effect from 51% of baseline to the present figure. A binary outcome and a regressor with a standard deviation of 0.07 do not support a tighter design. The $\text{Post} \times \text{WFH_Exposure}$ main effect, by contrast, is negative and significant in both specifications (-0.1555^{**} and -0.1516^{**}). This indicates a general post-2021 employment decline in high-exposure cells that is unrelated to motherhood specifically.

This is not for want of a working exposure measure. The cell-based index strongly predicts realized post-period WFH: a first-stage regression of the reference-week WFH indicator on exposure in the post-period yields 0.6557^{***} (SE 0.0954, $n = 110,051$), stable across 2021–2023 (interactions with 2022 and 2023 of 0.0118 and 0.0379, the latter marginal). Nor is it a specification failure: the exposure cells are deliberately finer than the fixed-effect cells, so the regressor retains within-cell variation (R^2 of exposure on the fixed-effect variables 0.285, VIF 1.40). An earlier version of the design, in which exposure was built on exactly the regression’s own control variables, had an MDE of 51% of baseline; successive refinements of the cell definition halved it to the current 26%, and cell-level aggregation was confirmed to add no power. The remaining limitation is structural: exposure assigned at the demographic-cell level simply does not vary enough. This is what motivates the intensive-margin analysis, where occupation-level exposure is available.

6.3 Intensive margin: hours

Pre-trends. Before turning to the estimates, Table 4 and Figure 6 report the hours event study. Relative to 2019, neither pre-period $\text{Mother} \times \text{year}$ coefficient is distinguishable from zero — 0.3515 in 2017 and 0.2013 in 2018 — and a joint Wald test on the two does not reject ($F(2, 63,198) = 0.949$, $p = 0.387$). The same test on the employment outcome also passes ($F(2, 79,069) = 0.63$, $p = 0.53$), so the difference-in-differences identifying assumption is supported on both margins and over the full pre-period. Both of these are tests of equation (1). The triple difference rests on a different and stricter assumption, which requires an event study of its own; that test is reported below in Table 7 and Figure 7. In the post-period the effect builds gradually — 0.3923 in 2021, 0.1213 in 2022, and 0.7517^{**} in 2023. Figure 6 plots the same coefficients with their confidence intervals, and Figure 1 shows the corresponding raw gaps with no regression adjustment at all.

The hours DiD: a null. Column (1) of Table 5 reports equation (1) for weekly hours among women working in the reference week. Mothers work 1.739 fewer hours per week than childless women ($\text{Mother} = -1.739^{***}$), the intensive-margin penalty itself. That

Table 3: Extensive-margin DDD: employment $\times$ cell-based WFH exposure

	Spec 1: additive controls	Spec 2: interacted cell FE
Mother $\times$ Post $\times$ WFH_Exposure	−0.0257 (0.0725)	−0.0194 (0.0722)
Mother $\times$ Post	0.0065 (0.0168)	0.0044 (0.0166)
Mother $\times$ WFH_Exposure	−0.1404 (0.0803)	−0.0818 (0.0702)
Post $\times$ WFH_Exposure	−0.1555** (0.0566)	−0.1516** (0.0554)
WFH_Exposure	0.0131 (0.0565)	0.0056 (0.0490)
MDE ($\alpha = 0.05$, power 0.80), per unit	0.2031	0.2022
MDE per SD of exposure	0.0143	0.0143
as % of baseline employment rate	1.85%	1.85%
MDE / point estimate	7.9 $\times$	10.4 $\times$
Observations	355,984	355,984
R^2	0.22103	0.22952

Dependent variable: employment indicator; full sample of employed and non-employed women. Both specifications include Mother $\times$ age-group interactions and the standard controls (Spec 2 absorbs age group, education, and district into interacted cell fixed effects). Standard errors clustered on the age-by-education-by-district cell. Baseline employment rate 0.7737. The MDE is a coefficient, so its per-unit value answers a question the data cannot pose: the exposure index has a standard deviation of 0.0706 and an interquartile range of 0.0766 on this sample, and never spans a unit. The per-SD row is the interpretable one. The ratio to the point estimate is invariant to scaling and is the basis for calling this null uninformative.

Table 4: Event study: Mother $\times$ year, weekly hours (reference 2019)

Year	Coefficient	SE
2017	0.3515	0.2570
2018	0.2013	0.2191
2021	0.3923	0.2612
2022	0.1213	0.2654
2023	0.7517**	0.2650
Joint test, 2017–2018	$F(2, 63, 198) = 0.949, p = 0.387$	
Observations	251,857	

Women working in the reference week; standard controls; standard errors clustered by individual. Neither pre-period coefficient is significant and the joint test does not reject.

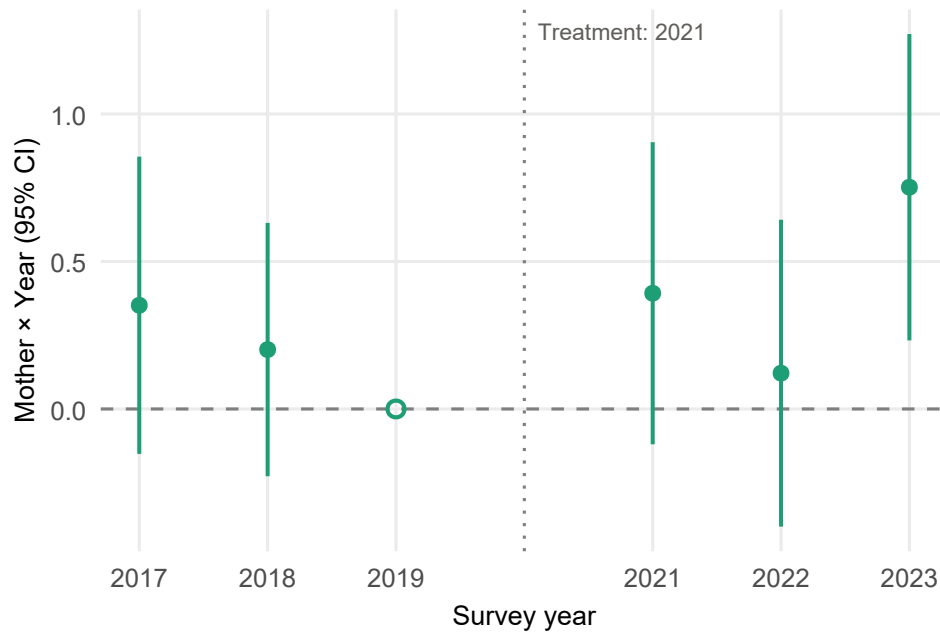


Figure 6: Event study: Mother $\times$ year interactions for weekly hours, reference year 2019

Notes: Women working in the reference week; standard controls; standard errors clustered by individual. Vertical bars are 95% confidence intervals. The hollow point at 2019 is the omitted reference year, pinned at zero by construction rather than estimated. The gap between 2019 and 2021 is the excluded 2020 survey year; the dotted rule marks the treatment boundary. Point estimates are those of Table 4. Neither pre-period coefficient differs significantly from zero.

penalty did not measurably change after the shift to remote work: $\text{Mother} \times \text{Post} = 0.2280$ (SE 0.1809), $t = 1.26$, $N = 251,857$. The raw cell means of Section 4.2 agree on magnitude: the unadjusted difference-in-differences is 0.266, so the controls move the estimate by less than four hundredths of an hour. They do not agree on precision — the unclustered standard error is 0.098, roughly half the clustered one — which is why the raw arithmetic should not be read as evidence of a significant increase.

Following the standard set in Section 6.1, it is worth saying what kind of null this is. At this standard error the design’s minimum detectable effect is $2.8016 \times 0.1809 \approx 0.51$ hours, so the estimate rules out average gains much above half an hour a week. This is a reasonably precise null, not an uninformative one — a distinction that matters here for the same reason it mattered on the extensive margin.

Selection into the estimation sample does not change that reading. The probability that hours are observed for post-period mothers (0.6964) exceeds the parallel-trends-in-selection counterfactual ($s_{11}^* = 0.6786 + (0.7058 - 0.7010) = 0.6833$), so there is excess selection to trim — about 1.9% of the cell.² Table 6 reports the resulting Lee bounds: the lower bound is -0.5166 (SE 0.1781) and the upper bound 0.8017 (SE 0.1789), with an Imbens–Manski 95% confidence interval for the identified set of $[-0.810, 1.096]$. The correction widens the interval around a null rather than overturning a finding. We therefore do not treat the plain hours DiD as evidence of an average effect in either direction, and nothing that follows rests on it.

The hours DDD: the paper’s headline result. Column (2) of Table 5 interacts the DiD with the calibrated occupation-level exposure index. The triple interaction is 3.224** (SE 1.022, $N = 246,326$): among mothers who are working, those in more WFH-exposed occupations saw a larger post-2021 increase in weekly hours relative to childless women in the same occupations — about 3.2 additional hours per unit of exposure, where exposure runs from zero to one.

The column’s second row is as informative as its first. At zero exposure — an occupation that cannot be done from home at all — the implied change for mothers is $\text{Mother} \times \text{Post} = -0.5385$ (SE 0.3364), if anything negative. Set against the near-zero average effect of column (1), the reading is not that a broad gain is *larger* in teleworkable jobs. It is that outside teleworkable jobs there is no gain to speak of, and the entire effect is the exposure gradient. That is a sharper claim than the one a significant average DiD would have supported, and it is the claim Figure 2 shows directly: three quartiles at zero and one well above it.

The estimate remains inside the design’s power, though by less of a margin than the extensive margin’s failure might suggest is easy: the minimum detectable effect is 2.8640

²Derived from the selection rates reported in the note to Table 6 ($1 - 0.6833/0.6964$), not directly exported by the pipeline.

Table 5: Intensive-margin DiD and DDD: weekly hours, reference-week workers

	(1) DiD	(2) DDD, calibrated exposure
Mother $\times$ Post $\times$ WFH_Exposure		3.224** (1.022)
Mother $\times$ Post	0.2280 (0.1809)	-0.5385 (0.3364)
Mother $\times$ WFH_Exposure		-1.295 (1.053)
Post $\times$ WFH_Exposure		-0.5483 (0.8102)
WFH_Exposure		1.128 (3.912)
Mother	-1.739*** (0.1441)	-1.449*** (0.3512)
Post	-0.0570 (0.1481)	0.0535 (0.2732)
Constant	32.35*** (0.3182)	32.12*** (1.540)
MDE for triple interaction		2.8640
Observations	251,857	246,326
R^2	0.04049	0.04094
Clustering	individual	2-digit occupation

Dependent variable: usual weekly work hours, women working in the reference week. Standard controls in both columns. Column (2) uses the calibrated occupation-level exposure index, and its sample is column (1)'s less the rows with no matched 2-digit occupation. MDE at $\alpha = 0.05$, power 0.80. Note that column (2)'s Mother $\times$ Post is the effect at *zero* exposure, not an average effect.

Table 6: Lee bounds on the hours DiD (Mother $\times$ Post)

Bound	Coefficient	SE	95% CI
Lower	-0.5166	0.1781	[-0.8657, -0.1675]
Point (untrimmed)	0.2280	0.1809	[-0.1266, 0.5825]
Upper	0.8017	0.1789	[0.4511, 1.1523]
Imbens–Manski 95% CI			[-0.810, 1.096]

Selection rates $s_{ab} = \Pr(\text{hours observed})$: $s_{00} = 0.7010$ ($n = 71,532$), $s_{01} = 0.7058$ (62,114), $s_{10} = 0.6786$ (129,856), $s_{11} = 0.6964$ (109,239). These are lower than the corresponding employment rates because they condition on working in the reference week as well as on holding a job.

hours (7.3% of mean weekly hours, or 88.8% of the point estimate). The contrast with the employment DDD — whose MDE is eight to ten times *its* point estimate — still holds, but this design detects an effect of the size observed without a great deal of room to spare.

Pre-trends for the triple difference. The pre-trend tests of Table 4 license equation (1), not equation (2). They ask whether mothers and childless women trended together; the triple difference requires the stricter condition that the mother/non-mother gap trended together *across levels of WFH exposure*. These are not the same assumption, and the difference-in-differences version is not the conservative one: because it averages over exposure, a pre-period divergence correlated with exposure but netting to zero across occupations would leave the test of Table 4 untouched while invalidating the headline estimate. We therefore estimate the event-study analogue of equation (2) — replacing Post with a full set of survey-year interactions at every level of the interaction, so that the triple difference is recovered separately for each year against 2019 — on the same sample, the same exposure index, and the same occupation-level clustering as column (2) of Table 5.

Table 7 and Figure 7 report the result, and it is supportive. Neither pre-period coefficient is close to significant, and neither is far from zero in level: 0.5866 in 2017 and 0.3134 in 2018, against post-period estimates five to thirteen times larger. The joint Wald test does not reject, and does not come near rejecting ($F(2, 39) = 0.041$, $p = 0.960$). The post-period path is flat, then steps up in 2021 and rises: 3.011*, 3.545**, 4.020**, all three individually significant and straddling the pooled 3.224 of Table 5, which is what a treatment beginning in 2021 looks like rather than a pre-existing trend carried through the sample.

One caveat belongs with this test rather than in a footnote. Clustering at the level the exposure regressor actually varies at leaves forty occupation clusters, so the pre-period standard errors (2.2367 and 1.7449) are of the same order as the effect being tested. Each pre-period confidence interval therefore contains values larger in magnitude than 3.224: the test is consistent with parallel trends and shows no sign of a violation, but it does not have the power to rule out a pre-trend as large as the estimate itself. This is the same small-cluster limitation the headline estimate carries, and we read it as supporting evidence rather than as a closed question.

Lee bounds for the DDD. Table 8 applies the generalized selection correction. Excess selection into the estimation sample among post-period mothers is present in all four quartiles of the cell-based exposure index but modest everywhere: the trim proportion ranges from 0.80% (quartile 3) to 4.48% (quartile 1, the least teleworkable jobs), and a total of 1,396 of 72,820 post-period working mothers are trimmed. The bounds move little: the lower bound is 3.1255 (SE 1.1440), the upper bound 3.4434 (SE 1.0220), and the Imbens–Manski 95% confidence interval for the identified set is [1.021, 5.324], which

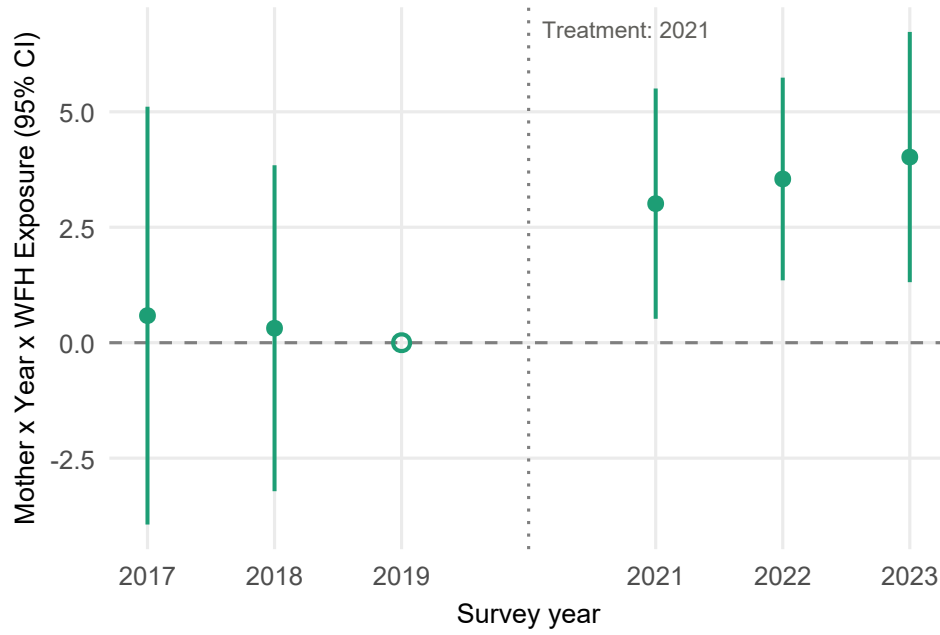


Figure 7: Event study for the triple difference: $\text{Mother} \times \text{year} \times \text{WFH_Exposure}$ interactions for weekly hours, reference year 2019

Notes: Point estimates of Table 7; vertical bars are 95% confidence intervals from occupation-clustered standard errors. The hollow point at 2019 is the omitted reference year, pinned at zero by construction rather than estimated. The gap between 2019 and 2021 is the excluded 2020 survey year; the dotted rule marks the treatment boundary. Compare Figure 6, which plots the difference-in-differences analogue and tests a weaker assumption.

Table 7: Event study: Mother $\times$ year $\times$ WFH_Exposure, weekly hours (reference 2019)

Year	Coefficient	SE
2017	0.5866	2.2367
2018	0.3134	1.7449
2021	3.0114*	1.2327
2022	3.5454**	1.0850
2023	4.0203**	1.3397
Joint test, 2017–2018	$F(2, 39) = 0.041, p = 0.960$	
Observations	246,326	

Employed women matched to a two-digit occupation and working in the reference week; standard controls; calibrated occupation-level exposure; standard errors clustered by two-digit occupation (40 clusters). The specification includes the year main effects and both two-way year interactions (Mother $\times$ year and WFH_Exposure $\times$ year) alongside the triple interaction reported here; omitting any of them would make the estimate a two-way difference rather than a triple one. Coefficients are hours per unit of exposure, where exposure runs from zero to one. Neither pre-period coefficient is significant and the joint test does not reject.

excludes zero. The triple interaction therefore survives the selection correction, and is the strongest single piece of evidence in the paper.

Robustness. Table 9 collects the remaining checks. With the raw [Dingel and Neiman \(2020\)](#) external index in place of the calibrated one, the triple interaction is 0.6720 (SE 0.8838), positive but insignificant; with the realized 2021-anchored index it is 5.195* (SE 2.363). The direction is consistent across all three measures but the magnitude and significance are not, and we describe the result as directionally rather than point-estimate robust to the exposure measure. The calibrated measure is the one purpose-built to correct the external index’s known misclassification of Israeli occupations, and we lead with it. Adding a Mother $\times$ age-group interaction leaves the estimate essentially unchanged (3.205**, SE 1.055); reweighting the sample to balance pre-period age groups between mothers and non-mothers attenuates it by about 19% to 2.597* (SE 1.050) — still significant, though at the 5% level rather than the 1%, and close enough to the design’s minimum detectable effect of 2.864 that we would not want to rest the result on this specification alone. The age imbalance being corrected is real and tracks exposure — the pre-period mother/non-mother gap in the age-group code is -0.974 in the lowest exposure quartile ($t = -94.0$), -0.735 in the second, -0.169 in the third, and $+0.057$ in the highest — so the reweighted attenuation is worth reporting, and it is the largest single move any

Table 8: Generalized Lee bounds on the hours DDD (Mother $\times$ Post $\times$ WFH_Exposure)

<i>Panel A: selection rates and trimming by cell-exposure quartile</i>							
Quartile	s_{00}	s_{01}	s_{10}	s_{11}	s_{11}^*	$n(\text{Mother}=1, \text{Post}=1)$	Trim
1	0.5830	0.5862	0.5497	0.5788	0.5529	21,655	4.48%
2	0.7070	0.7120	0.6856	0.6977	0.6906	28,540	1.01%
3	0.7542	0.7450	0.7554	0.7522	0.7462	34,293	0.80%
4	0.8194	0.8029	0.7352	0.7399	0.7188	21,815	2.85%
<i>Panel B: bounds on the triple interaction</i>							
Bound	Coefficient		SE		95% CI		
Lower	3.1255		1.1440		[0.8833, 5.3677]		
Point (untrimmed)	3.2319		1.0538		[1.1664, 5.2973]		
Upper	3.4434		1.0220		[1.4404, 5.4465]		
Imbens–Manski 95% CI				[1.021, 5.324]			

s_{ab} is the probability that hours are observed — employed *and* working in the reference week — so these rates are lower than the corresponding employment rates. Panel A’s n column counts *all* post-period mothers in the quartile, working or not, because the cell-based exposure index is defined for the non-employed as well. The trim proportions apply to the occupation-matched subset with observed hours, which is the population the bound regressions use: 548 of 12,249; 197 of 19,511; 202 of 25,317; 449 of 15,743, or 1,396 of 72,820 in total. The untrimmed point estimate here (3.2319) differs slightly from Table 5 (3.224) because this sample additionally requires a matched cell-based exposure. Outcome regressions use the occupation-level index and cluster by occupation.

check here produces. An incidental finding of the age-interacted specification is a rising age profile in the hours penalty itself, independent of exposure (Mother $\times$ age group 4: -2.266^{***} ; group 5: -1.150^* ; groups 6 and 7 positive but not significant at conventional levels).

Table 9: Hours DDD: sensitivity of Mother $\times$ Post $\times$ WFH_Exposure

Specification	Coefficient	SE	N
<i>Exposure measure</i>			
Calibrated (primary)	3.224**	1.022	246,326
External (Dingel–Neiman)	0.6720	0.8838	246,326
Realized (2021 anchor)	5.195*	2.363	246,037
<i>Age-balance</i>			
Age-interacted (+ Mother $\times$ age group)	3.205**	1.055	246,326
Rewighted (pre-period age-group raking)	2.597*	1.050	242,552

All rows: weekly hours, women working in the reference week, standard controls, clustered by occupation.

6.4 Heterogeneity by population group and the gender placebo

Table 10 re-estimates the hours DiD and DDD separately for Jewish and Arab women, and on men as a placebo. For Jewish women the pattern is the pooled pattern: the DiD is 0.3021 (SE 0.2027), indistinguishable from zero, while the DDD is 2.8465** (SE 0.9125), with a confidence interval that excludes zero. For Arab women the DiD is -0.0724 (SE 0.4785), also indistinguishable from zero — and Arab women’s hours fell in the post-period overall (Post = -2.162^{***}) — while the DDD is 5.5069^* (SE 2.6741, 95% CI [0.2657, 10.7481]), nominally significant and nearly twice the Jewish point estimate. Neither group shows an average effect; both show an exposure gradient.

We do not read the larger Arab estimate as evidence that the mechanism is stronger for Arab women, for two reasons. First, the two coefficients are not statistically distinguishable. The Arab subsample is about nine times smaller than the Jewish one (24,307 versus 209,727 occupation-matched women with observed hours), the Arab standard error is nearly three times the Jewish one, and a two-sample z -test on the DDD term gives $z = (5.5069 - 2.8465) / \sqrt{2.6741^2 + 0.9125^2} = 0.942$, $p = 0.346$ — the apparent gap is well within what the Arab subsample’s own sampling noise can produce. (The same test on the DiD gives $z = -0.720$, $p = 0.471$.) Second, there is a plausible mechanical cause for the imprecision: Arab women’s employment in Israel is concentrated in a narrow set of occupations, disproportionately education and health-aide roles, so that comparatively few two-digit occupations carry enough Arab respondents to inform an occupation-specific comparison. With few occupations carrying real information, a small number of high-leverage occupations can dominate the estimate. The Arab result is therefore best described as imprecisely estimated and statistically indistinguishable from the Jewish estimate — a small-sample, thin-occupational-coverage fragility rather than a documented heterogeneous treatment effect. The Jewish-women estimate is the more credible of the two and anchors any subgroup claim we make.

The gender placebo has to be read separately on the two margins, because it behaves differently on each. Re-estimating the same models on men, with “mother” read as “father,” gives a DiD of -0.4002^* (SE 0.1883, $N = 261,543$) and a DDD of -1.7620 (SE 1.0153, $N = 251,779$), the latter significant only at the 10% level and negative.

On the triple interaction the placebo does what a placebo should, and more. Fathers show no positive WFH-linked hours response; their point estimate is negative; and the women’s and men’s coefficients are statistically distinguishable ($z = 3.461$, $p = 0.0005$). That is a stronger statement than “the placebo is insignificant,” because a macro trend common to all workers cannot produce a positive exposure gradient for mothers alongside a negative one for fathers. The mechanism claim rests on this margin.

On the plain DiD the placebo is not merely uninformative but adverse, and we would rather say so than let a reader discover it. Fathers’ hours fall by 0.40 (SE 0.19) relative

to childless men — a *larger* movement, and a more statistically significant one, than the 0.23 (SE 0.18) we estimate for mothers. The two are still distinguishable ($z = 2.406$, $p = 0.016$), but the difference is driven by the fathers’ coefficient rather than the mothers’. A design whose placebo moves more than its treatment group has not demonstrated a clean average effect on that margin, and we do not claim one: Section 6.3 reads the female DiD as a null, and the placebo is consistent with that reading rather than against it. What would be troubling is a significant *positive* placebo on the exposure gradient, and there is none.

Table 10: Hours DiD and DDD by subgroup, and the gender placebo

Subgroup	DiD: Mother $\times$ Post			DDD: Mother $\times$ Post $\times$ WFH_Exposure		
	Coef.	SE	N	Coef.	SE	N
All women (primary)	0.2280	0.1809	251,857	3.2240**	1.0223	246,326
Jewish women	0.3021	0.2027	214,449	2.8465**	0.9125	209,727
Arab women	−0.0724	0.4785	24,975	5.5069*	2.6741	24,307
Men (placebo)	−0.4002*	0.1883	261,543	−1.7620	1.0153	251,779
95% CI, Jewish	[−0.0952, 0.6994]			[1.0579, 4.6351]		
95% CI, Arab	[−1.0103, 0.8656]			[0.2657, 10.7481]		
Jewish vs. Arab, z (p)	−0.720 (0.471)			0.942 (0.346)		
Women vs. men, z (p)	2.406 (0.016)			3.461 (0.0005)		

Weekly hours, women working in the reference week, standard controls. DiD clustered by individual; DDD clustered by occupation. * denotes significance at the 10% level. Men: “Mother” read as “Father.” Two-sample z -tests are valid without a covariance correction because the compared samples are disjoint.

7 Discussion

Among mothers who remain in work, there is no post-2021 hours effect on average. There is a large one in teleworkable occupations. Those two statements are the paper’s result, and it is the conjunction that makes it informative: a triple-interaction coefficient of about 3.2 additional weekly hours per unit of exposure, an implied change at zero exposure that is if anything negative, and a selection correction the estimate survives. This is consistent with WFH access relaxing a binding time constraint specifically for mothers in teleworkable jobs — the Goldin (2014) mechanism — and it is difficult to reconcile with a general post-pandemic recovery common to all women, which would have raised hours across occupations rather than only where the work can be done from home. Fathers show no analogous exposure gradient.

Which occupations are doing the work is concrete. The calibration step moved ten occupations away from their US task-based scores toward realized Israeli behavior: clerical support and teaching were corrected sharply downward, because Israeli institutional

practice kept them in person regardless of task content, while ICT professionals remained high on both measures. The triple interaction is identified off exactly this contrast, which grounds “exposure” in checkable occupational content. That the raw external index yields a positive but insignificant estimate is itself informative: the mechanism shows up cleanly only once exposure is measured as Israeli jobs were actually done, not as US task content says they could be.

The timing is also informative. The event study rises gradually from an insignificant 2021 estimate to a significant one in 2023, consistent with WFH adoption and household time reallocation being a multi-year adjustment rather than an instantaneous response to the 2020–21 shock — plausible given that occupational WFH norms were themselves still settling over 2021–2023.

Why hours and not employment? The employment DDD is not a failed test of the mechanism; it is an uninformative one: its minimum detectable effect is roughly eight to ten times its own point estimate, and about twice to three times the effect the closest study in the literature reports. The intensive-margin design is well powered because it can use occupation-level exposure, which the employment design cannot without conditioning its own outcome on itself. Substantively, the two results are coherent: the effect operates on *how much* mothers in teleworkable jobs work, not on *whether* they are employed, which is precisely the margin [Kleven et al. \(2019b\)](#) identify as a leading channel of the child penalty. This is also where our findings depart from [Harrington and Kahn \(2025\)](#): their US evidence is on employment and income, and we cannot confirm or reject an employment effect in Israel with this design; what we can say is that the intensive margin responds to *remote-work exposure*, and not otherwise. A purely statistical account is not mutually exclusive, and on the present results it is the least strained one: the continuous hours outcome carries strictly more information per observation than a binary employment indicator and is correspondingly better powered to detect an effect of realistic magnitude, so the contrast between the two margins may be as much about what each design can see as about what workers did.

We previously read this pattern as evidence that labor supply adjusts on the cheaper margin — hours — before the costlier one of leaving employment, an account for which Israel’s furlough (*Halat*) system offers institutional support. We no longer think the data bear that reading. A margin-ordering story predicts an average hours response, and there is none; and it says nothing about why the response we do find is graded by teleworkability. The Goldin time-constraint mechanism accounts for the exposure gradient directly, and we rest on it. The policy reading is correspondingly narrower: WFH availability matters for how intensively mothers who already hold *teleworkable* jobs can work, not for whether mothers in general hold jobs or how much they work — which is where our evidence speaks, and where it does not.

The subgroup results are more modest than the point estimates suggest. The Jewish-

women estimate carries the pooled result. The Arab-women estimate is larger but imprecise, statistically indistinguishable from the Jewish one, and rests on thin occupational coverage; we report it transparently but do not claim heterogeneity on its basis.

8 Limitations

Parallel trends, and a measurement correction that produced them. Joint pre-trend tests do not reject on either margin — $F(2, 63, 198) = 0.949$, $p = 0.387$ for hours and $F(2, 79, 069) = 0.63$, $p = 0.53$ for employment — so the identifying assumption is supported over the full 2017–2019 pre-period rather than a truncated version of it. The triple difference’s own pre-trend test, on the event-study version of equation (2) (Table 7), does not reject either ($F(2, 39) = 0.041$, $p = 0.960$). That test is the one that matters for the headline estimate, and it is also the weakest of the three: with forty occupation clusters its pre-period standard errors are of the same order as the effect being tested, so it cannot exclude a pre-trend the size of the estimate. It is evidence of no visible violation, not proof of none.

That is worth stating carefully, because it is true only after a correction. As described in Section 3, the 2017 file records zero usual hours for respondents who were employed but did not work in the reference week, a convention no later year follows. Since absence from work is more common among mothers, those zeros depressed mothers’ measured 2017 hours specifically, and an earlier version of this analysis read the result as a significant pre-trend violation confined to 2017 — large enough that the identifying assumption had to be narrowed to 2018–2019. Defining hours on reference-week workers in every year removes the artifact and the apparent violation with it. We report this rather than silently presenting a passing test: the sensitivity of a parallel-trends diagnostic to a single year’s coding convention is itself worth knowing, and a reader comparing these figures against an earlier draft is entitled to know why they differ.

The correction is not costless. The estimand is now usual hours among women who were working, not among all women holding a job, and roughly 10% of each year’s employed sample is excluded on that basis. If absence from work in the reference week is itself related to WFH exposure — for instance if teleworkable jobs make it easier to work while unwell or while caring for a child — then the restriction is not innocuous for the triple interaction. We find the absentee share only mildly related to exposure across quartiles, but we cannot rule this out.

Lee bounds: direction and monotonicity. The trimming construction only handles *excess* selection in the post-period mother cell — the direction the project’s own hypothesis predicts. Under-selection in that cell (fewer employed mothers than the counterfactual implies) is a missing-data problem, not an excess-observed-data problem, and is not ad-

dressed. The construction assumes monotone selection (that WFH weakly increases, never decreases, a mother’s employment probability) to justify trimming from a single tail, and computes selection rates unweighted, consistent with the rest of the analysis. A finding of small excess selection should be read as “this construction found little to correct,” not as proof that the selection concern is absent.

Survey weights. CBS survey weights are intentionally not applied in any outcome regression. This is a scope decision, not an oversight: reported rates and coefficients are unweighted estimates on the analysis sample, not population-representative statistics. The one exception is internal to constructing the cell-based exposure index, where occupation exposure is aggregated to demographic cells using the survey weight; that is not a representativeness correction for the outcome regressions.

The exposure anchor: 2021, not 2020. The realized WFH index is anchored to 2021 rather than to the 2020 variable the original research plan specified, because no 2020 extract exists. The justification is that 2021 already passes the validated data pipeline and is arguably a more stable measurement point than 2020’s emergency disruption. The downside is real: by 2021 occupations had had a year to adapt, and if adaptation was uneven a 2021-based index could rank occupations’ exposure differently than a 2020-based one would have. The realized index enters only as a robustness variant; the primary calibrated index draws on 2022–23 realized data for its swap test and is itself a documented compromise rather than a fully pre-treatment measure.

Age controls and age imbalance. Age enters as a categorical age-group code, because the extract has no continuous age or birth-year variable. Mothers and non-mothers are imbalanced on this code in the pre-period, and the imbalance tracks exposure quartile; the age-interacted and reweighted specifications address this and the estimate survives, but the reweighted attenuation ($3.224 \rightarrow 2.597$) is the largest single move any robustness check produces, and it leaves the estimate only narrowly above the design’s minimum detectable effect of 2.864. The result should be read as robust to age imbalance in direction and approximate magnitude, not as insensitive to it.

Inference. The hours DDD clusters on about 40 occupation groups and the employment DDD on about 210 demographic cells; no small-cluster correction is applied. Clustering on the individual matters a good deal more than it may appear: the survey is a rotating panel, and the hours estimation sample of 258,175 observations comes from only 64,602 distinct respondents — about four observations each, and two to three within a single survey year, with 96% of rows belonging to someone who appears more than once. Treating those observations as independent would understate standard errors by a factor of roughly 1.8,

which is why every interval in this paper, including the descriptive ones in Section 4, is clustered. What the panel structure does *not* support is within-person identification: an individual-level fixed-effects design was considered and rejected because only 1.9% of respondents are re-interviewed across the pre/post boundary, so almost none of the repetition spans the treatment. Occupation codes masked by the CBS for disclosure control are dropped when building exposure; a proxy check finds no strong evidence that this biases the exposure measure, though that is weak rather than conclusive evidence. The calibrated exposure measure uses post-period (2022–23) realized data, as noted.

9 Conclusion

[SKELETON ONLY — REPLACE WITH YOUR OWN TEXT BEFORE SUBMISSION]

1. **Restate the question and the move.** [Harrington and Kahn \(2025\)](#) for the US; this paper takes it to Israel and to the intensive margin. Not a replication. (Section 2.)
2. **Extensive margin: two different nulls, do not conflate them.** The employment DiD is a precise null (-0.0053 , SE 0.0052). The employment DDD is uninformative, not null: MDE 0.2031 per unit of exposure, which is 1.43 percentage points per standard deviation and $8\text{--}10$ times the point estimate (Sections 6.1–6.2).
3. **Intensive margin: no average effect.** The plain hours DiD is 0.2280 (SE 0.1809), a reasonably precise null — it rules out average gains above about half an hour a week — and the selection correction widens rather than overturns it (Imbens–Manski $[-0.810, 1.096]$). Report it as a null, not as a weak finding (Section 6.3).
4. **The actual result.** Triple interaction 3.224 (SE 1.022 , $p < 0.01$), which does survive the generalized correction: Imbens–Manski $[1.021, 5.324]$. Inside the design’s power, though not by a wide margin (MDE 2.864 , or 89% of the estimate). This is the paper’s claim (Section 6.3).
5. **Why it reads as a mechanism and not a trend.** There is no average effect to decompose. At zero exposure the implied change is -0.5385 ; the entire effect is the exposure gradient, and the dose-response shows three quartiles at zero and one well above it. The effect is not a uniform post-2021 shift; it is absent outside teleworkable jobs. Link to [Goldin \(2014\)](#) (Section 7).
6. **Measurement is part of the result.** The raw [Dingel and Neiman \(2020\)](#) index gives 0.6720 (insignificant); the Israeli-calibrated index gives 3.224 . The mechanism is visible only once exposure is measured as Israeli jobs were actually done (Section 7).
7. **The placebo, split by margin.** On the DDD it supports the claim: fathers -1.7620 , opposite sign, women versus men $z = 3.461$ ($p = 0.0005$) — a common macro trend

cannot produce a positive exposure gradient for mothers and a negative one for fathers. On the plain DiD it is adverse and should be conceded: fathers move -0.4002^* , more than the 0.2280 we estimate for mothers. Consistent with reading the female DiD as a null (Section 6.4).

8. **Be modest about subgroups.** Jewish women (2.8465) carry the pooled result. The larger Arab estimate (5.5069) is statistically indistinguishable from it ($z = 0.942$, $p = 0.346$) and rests on thin occupational coverage. Do not claim heterogeneity.
9. **Own the limitations, do not bury them.** Parallel trends now hold on both margins, but only after correcting a 2017 coding convention that recorded zero usual hours for the employed-but-absent — disclose the correction and the narrowed estimand it implies; the bounds handle only excess selection and assume monotonicity; the exposure index is anchored to 2021 and the calibration draws on 2022–23; estimates are unweighted; inference rests on about 40 occupation clusters with no small-cluster correction (Section 8).
10. **Close on what it means and what is next.** Policy reading: WFH availability bears on how intensively mothers already employed can work, not on whether they are employed. Next steps: a higher-powered test of the employment margin, and a jointly estimated model letting the mechanism differ by sex and population group.

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